

Transformations & Constructions

Quick reference sheet · GCSE Maths, Geometry & Measures · keep handy for revision

1. Translation

A translation slides a shape — no turning, flipping or resizing. Described by a column vector (x on top = left/right, y on bottom = up/down). To translate a point, add the vector to its coordinates: $(x, y) + (a, b) = (x + a, y + b)$.

2. Rotation

A rotation turns a shape about a fixed point. Must state THREE things: angle, direction (clockwise/anticlockwise), and centre of rotation. About the origin: 90° clockwise $(x, y) \rightarrow (y, -x)$; 90° anticlockwise $(x, y) \rightarrow (-y, x)$; 180° $(x, y) \rightarrow (-x, -y)$.

3. Reflection

Mirror line	Effect on point (x, y)
x-axis	$(x, -y)$
y-axis	$(-x, y)$
$y = x$	(y, x)
$y = -x$	$(-y, -x)$
$x = a$ (vertical line)	Same distance from $x = a$, on the other side

4. Enlargement

Described by a scale factor (SF) and a centre of enlargement. Each image point lies on the ray from the centre through the object point, at $SF \times$ the distance from the centre. **Area scale factor = (linear SF)².**

Scale factor	Size of image	Position vs. centre
$SF > 1$	Bigger	Same side as object
$0 < SF < 1$	Smaller	Same side as object
$SF < 0$	$ SF \times$ object size	Opposite side, turned 180°

5. Constructions (ruler and compasses only)

Perpendicular bisector of AB: compasses set to more than half of AB; arcs from A and from B (same radius) crossing above and below the line; join the crossing points. Result cuts AB at 90° through its midpoint.

Angle bisector: arc from the vertex crossing both arms; from each of those points, equal-radius arcs crossing inside the angle; join the vertex to the crossing point. Result splits the angle exactly in half.

Perpendicular from a point to a line (point not on the line): arc from the point crossing the line twice; equal-radius arcs from those two crossing points, meeting on the far side; join the point to that meeting point.

Perpendicular at a point on a line: mark two points on the line equidistant from the given point; equal-radius arcs from those two marks, meeting above (or below) the line; join the given point to that meeting point.

Never rub out construction arcs — they are part of the method and earn marks.

6. Common mistakes

Describing a rotation without all three parts (angle, direction, centre).

Forgetting that a scale factor between 0 and 1 is still an enlargement, even though the shape gets smaller.

Placing the image on the wrong side of the centre for a negative scale factor.

Rubbing out compass arcs on a construction, or using a protractor instead of compasses.

Mixing up the top and bottom numbers in a column vector (top = left/right, bottom = up/down).